

HTML Full Cheat Sheet (All Concepts)

Complete guide with examples • Beginner → Advanced • Full Stack focused

1. HTML Introduction

HTML (HyperText Markup Language) defines the structure of the web page. It is made of elements written using tags. Tags describe meaning (semantic) and structure. HTML works together with: • CSS → styling (UI/UX) • JavaScript → interactivity and logic

```
<!-- HTML element -->
<p>Hello World</p>
```

2. HTML Document Boilerplate

Every HTML page should have a valid boilerplate. It helps browsers render correctly, improves SEO, and ensures responsive support.

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Document</title>
</head>
<body>
  <!-- visible content -->
</body>
</html>
```

3. Elements, Tags & Attributes

• Elements can be block or inline. • Attributes provide extra info (id, class, href, src, alt, type, name, value etc.). • Some tags are void/self-closing (img, br, hr, meta, link, input).

```
<a href="https://example.com" target="_blank" rel="noopener noreferrer">
  Visit
</a>



<input type="text" name="username" />
```

4. Headings & Text Formatting

Headings define hierarchy (SEO friendly). Use formatting tags for meaning: • strong/em (semantic) • mark, small, del, ins, sub, sup

```
<h1>Main Title</h1>
<h2>Section</h2>
<p>This is <strong>important</strong> and this is <em>emphasis</em>.</p>
<p>2<sup>2</sup>=4 and H<sub>2</sub>O</p>
<p><mark>Highlighted</mark> text</p>
```

5. Links (Anchor tag)

Links are navigation core. Common types: • external links • internal links • anchor jump (#id) • email/phone links

```
<a href="/about.html">About</a>
<a href="#contact">Jump to contact</a>
<a href="mailto:nikhil@gmail.com">Email</a>
<a href="tel:+919999999999">Call</a>

<section id="contact">...</section>
```

6. Images & Media

Always provide alt text. Use figure + figcaption for captions. Use video/audio for multimedia.

```
<figure>
  
  <figcaption>QuickBite Food Banner</figcaption>
</figure>

<video controls width="400">
  <source src="video.mp4" type="video/mp4" />
</video>

<audio controls>
  <source src="music.mp3" type="audio/mp3" />
</audio>
```

7. Lists

List types: • ul → unordered • ol → ordered • dl → description list

```
<ul>
  <li>Burger</li>
  <li>Pizza</li>
</ul>

<ol>
  <li>Sign up</li>
  <li>Login</li>
</ol>

<dl>
  <dt>API</dt>
  <dd>Bridge between frontend and backend</dd>
</dl>
```

8. Tables

Tables display structured data. Use: • thead/tbody/tfoot for better structure • scope attribute for accessibility

```
<table>
  <thead>
    <tr>
      <th scope="col">Name</th>
      <th scope="col">Role</th>
    </tr>
  </thead>

  <tbody>
    <tr>
      <td>Nikhil</td>
      <td>Full Stack Learner</td>
    </tr>
  </tbody>
</table>
```

9. Forms (Full Stack Must)

Forms send data to backend. Important attributes: • action, method • name attribute is key for backend Validation: • required, minlength, maxlength, pattern Input types: text, email, password, number, date, file, checkbox, radio

```
<form action="/register" method="POST">
  <label for="name">Name</label>
  <input id="name" name="name" type="text" required />
```

```

<label for="email">Email</label>
<input id="email" name="email" type="email" required />

<label for="pass">Password</label>
<input id="pass" name="password" type="password" minlength="6" required />

<button type="submit">Create Account</button>
</form>

```

10. Form Controls (Advanced)

Other form elements: • textarea • select + option • datalist • fieldset + legend

```

<textarea name="message" rows="4" placeholder="Write message..."></textarea>

<select name="city">
  <option value="delhi">Delhi</option>
  <option value="lucknow">Lucknow</option>
</select>

<input list="skills" name="skill" />
<datalist id="skills">
  <option value="HTML" />
  <option value="CSS" />
  <option value="JavaScript" />
</datalist>

<fieldset>
  <legend>Profile</legend>
  <input name="age" type="number" />
</fieldset>

```

11. Semantic HTML (SEO + Accessibility)

Semantic tags help screen readers and SEO. Use: header, nav, main, section, article, aside, footer

```

<header>
  <nav>
    <a href="/">Home</a>
    <a href="/projects">Projects</a>
  </nav>
</header>

<main>
  <section>
    <article>
      <h2>Blog Post</h2>
      <p>Content...</p>
    </article>
  </section>
</main>

<footer>© 2026 Nikhil</footer>

```

12. HTML Meta Tags

Meta tags are placed inside head. Important: • charset • viewport (mobile responsive) • description (SEO) • Open Graph for social sharing

```

<meta charset="UTF-8" />
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
<meta name="description" content="Portfolio of Nikhil - Full Stack Developer" />

<!-- Open Graph -->
<meta property="og:title" content="Nikhil Portfolio" />
<meta property="og:description" content="Full Stack projects and skills" />
<meta property="og:image" content="https://site.com/preview.png" />

```

13. HTML Accessibility (A11y)

Accessibility makes websites usable for everyone. Key points: • alt text for images • labels for inputs • proper heading order • aria-label for icons/buttons

```
<button aria-label="Close Modal">X</button>
```

```
<label for="phone">Phone</label>
<input id="phone" name="phone" type="tel" />
```

14. HTML Performance Best Practices

• compress images, use proper sizes • avoid heavy inline HTML • use lazy loading for images • use defer for scripts

```

```

```
<script src="app.js" defer></script>
```

15. SEO Essentials

SEO basics using HTML: • 1 h1 tag • proper title + meta description • semantic structure • descriptive links • use canonical in large sites

```
<title>Food Delivery Website | QuickBite</title>
<meta name="description" content="Order food online fast with QuickBite." />
```

```
<a href="/menu">View Menu</a>
```

16. HTML Entities

Entities are used for special characters.

```
&lt;   &gt;   &amp;   &quot;   &copy;   &nbsp;
```

17. iframe & Embeds

Used to embed maps/videos/other pages. Use with caution (performance & security).

```
<iframe
  src="https://www.youtube.com/embed/VIDEO_ID"
  title="YouTube video"
  width="560"
  height="315"
  allowfullscreen>
</iframe>
```

18. HTML Comments

Comments are ignored by browser but helpful for devs.

```
<!-- This is a comment -->
```

19. Best HTML Folder/Project Structure

For clean full stack frontend structure:

```
/project
  /assets
    /images
    /icons
  /css
    style.css
  /js
    app.js
  index.html
```

20. Mini Practice Tasks (Daily)

Use these tasks to become expert: 1) Build a login form with validation 2) Create a product table 3) Create a page layout using semantic tags 4) Make a contact form 5) Use meta tags + Open Graph

```
<!-- Practice: Create a login page with semantic layout -->
<header>...</header>
<main>
  <form>...</form>
</main>
<footer>...</footer>
```

End of Cheat Sheet — Created for Full Stack Learning